AFSIFormer: Adaptive Frequency–Spatial Interaction Attention Mechanism for Aerial Image Semantic Segmentation

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***Abstract*—Aerial image semantic segmentation continues to face significant challenges in accurately capturing boundary textures. While convolutional neural networks (CNNs) and transformers are effective at modeling local features and long-range contextual dependencies, they often struggle with fine-grained boundary representation. In contrast, frequency-domain information offers complementary advantages by effectively representing periodic textures and structural edges. In this article, we propose a novel adaptive frequency–spatial interaction transformer (AFSIFormer) that follows a progressive learning strategy. First, a boundary-aware directional attention mechanism (BADAM) captures long-range dependencies across windows. Then, a local window attention mechanism (LWAM) refines contextual information within each window, enabling fine-grained local modeling under global guidance. Both BADAM and LWAM are built upon our designed adaptive frequency–spatial interaction attention (AFSIAttention) to capture contextual information across different windows. Unlike the existing frequency- and spatial-domain external coarse integration strategies, this mechanism utilizes a head-specific lightweight frequency projection network (HS-LFPN) to dynamically generate frequency-domain weights for each attention head. These frequency weights interact adaptively with spatial attention (SpatAttn) weights, facilitating frequency-guided spatial feature learning and internal integration of frequency and spatial information. Furthermore, we design a block-level residual coupling architecture that embeds AFSIFormer as a residual module within each convolutional stage, allowing continuous infusion of global and frequency-domain cues throughout the network. These collectively constitute the synergistic frequency–spatial network (SynFSNet), which achieves the state-of-the-art (SOTA) performance on three benchmark aerial image segmentation datasets. The code is available at https://github.com/Xinmu-Tantai/SynFSNet**

***Index Terms*—Aerial image segmentation, attention mechanism, frequency–spatial interaction, remote sensing, residual coupling architecture.**

I. Introduction

SEMANTIC segmentation of aerial images enables precise pixel-level classification and plays a crucial role in land cover analysis [1], [2], disaster assessment [3], [4], environmental protection [5], road extraction [6], [7], and urban planning [8], [9]. However, high-resolution aerial images present unique challenges, as shown in Fig. 1. These include substantial intraclass appearance variations, a low proportion of pixels belonging to small objects, complex background interference, large-scale variations, and blurred object boundaries. These factors lead to high intraclass variance and low interclass separability [10], making boundary information difficult to extract accurately, which makes precise segmentation a continuous and challenging task.

In recent years, integrating frequency- and spatial-domain information has emerged as a promising approach to address segmentation boundary extraction challenges in aerial images. Frequency-domain analysis [11], [12] reveals structural features and texture patterns that conventional spatial-domain methods struggle to capture, playing a crucial role in understanding complex semantic relationships within aerial scenes. Signal processing theory [13], [14] shows that spatial and frequency domains emphasize distinct aspects of image representation: spatial domain emphasizes pixel distribution and local relationships, while frequency domain captures global periodic patterns and structural organization. The discrete cosine transform (DCT) [15] decomposes images into multiscale frequency components, where low-frequency components encode structural information and high-frequency components preserve edge details. This characteristic makes frequency-domain features particularly effective for handling complex boundaries in aerial imagery. In practice, regular farmland grids manifest as stable low-frequency distributions [16], while building outlines and road networks appear in high-frequency components [17]. DCT facilitates structural information separation from texture noise, enhancing model robustness against background interference. Furthermore, the orthogonality of DCT basis functions

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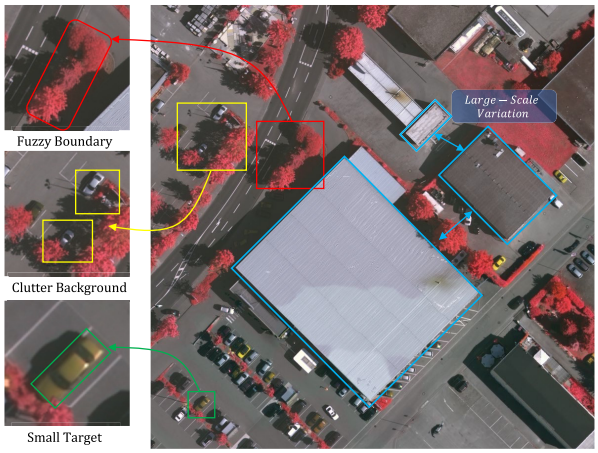


Fig. 1. Aerial imagery presents several prominent challenges. Buildings exhibit significant scale variations; tree boundaries are often blurred and difficult to distinguish from the surrounding environment, and cars are typically small, occupying only a few pixels and embedded in complex backgrounds, which increases the difficulty of accurate recognition.

reduces information redundancy, preserves frequency decomposition integrity, and enables more accurate identification of small-scale and blurred-boundary objects.

Several studies [18], [19], [20], [21], [22], [23], [24], [25] have explored integrating frequency- and spatial-domain information into semantic segmentation frameworks. For instance, SFFNet [26] employs independent branches to extract wavelet-transform features and fuse them with spatial-domain features, enhancing the overall representation. The spatial frequency network [27] introduces a frequency-domain weighting module to improve fusion quality. MIFNet [28] proposes a module that unifies local, global, and frequency information into a single representation. Although these methods ([18], [19], [21], [26]) have achieved notable performance improvements, their approaches to frequency–spatial-domain integration generally rely on external integration strategies—introducing frequency-domain information only after spatial-domain modeling is completed, with frequency–spatial interaction merely serving as a form of postprocessing. Furthermore, these integration strategies typically adopt a fixed 1:1 fusion ratio at the feature map or channel level. Such coarse integration presents several key limitations. First, it lacks the ability to act precisely on tokens, spatial positions, or structural units, thereby failing to enable truly fine-grained control. The uniform integration ratio also overlooks the varying requirements across different network stages. Second, as a postprocessing operation, the frequency–spatial interaction treats frequency information merely as a late-stage feature enhancement, making it difficult to serve as a core mechanism regulating the model's representational learning. The fundamental issue with this design lies in treating frequency-domain information merely as a simple enhancement to spatial-domain features, rather than as a mechanism for their guidance and regulation.

We further analyze the current mainstream aerial image semantic segmentation network architectures. Although convolutional neural networks (CNNs) excel at extracting local features, they struggle to model global contextual information due to their limited receptive field [29]. In contrast, transformers achieve long-range dependency modeling through self-attention mechanisms, demonstrating significant advantages in global representation capability [30]. Meanwhile, frequency-domain information possesses natural advantages in capturing periodic textures and structural boundaries in images and has been gradually introduced into aerial semantic segmentation tasks in recent years. However, the existing methods [31], [32], [33], [34], [35] often employ redundant dual-branch architectures and design complex fusion modules to combine frequency domain and spatial information. Combined with the inherently high computational complexity of transformers, the modeling of frequency domain or global information is typically limited to specific stages, such as the encoder, decoder, or deeper network layers [31], [32], [33], [34], [35], [36], [37], [38], as illustrated in Fig. 2(a)–(f). This stagewise modeling strategy leads to the weakening or even loss of frequency-domain features or global information during extraction and propagation processes, thereby limiting their practical effectiveness in remote sensing semantic segmentation.

To solve the above problems, we propose the AFSIAttention mechanism, design the adaptive frequency–spatial interaction transformer (AFSIFormer) block based on this mechanism, and construct an efficient semantic segmentation network synergistic frequency–spatial network (SynFSNet). Unlike the existing approaches that treat frequency-domain information as external and integrate it after spatial-domain modeling, AFSIAttention achieves deep interaction between frequency and spatial information within the attention mechanism itself, allowing frequency features to directly participate in the computation of attention weights, and models the contextual information in different windows through the features after frequency–spatial-domain interaction. AFSIFormer adopts a progressive modeling strategy by sequentially applying boundary-aware directional attention mechanism (BADAM) and local window attention mechanism (LWAM), achieving coarse-to-fine global information modeling while maintaining computational efficiency. Building on this, SynFSNet integrates a block-level residual coupling architecture [see Fig. 2(g)] and a bidimensional semantic enhancement module (BiSEM), ensuring continuous propagation of frequency and spatial information throughout the network, alleviating semantic information degradation, and significantly reducing the redundant information in the fusion of features at different levels through BiSEM. The main contributions of this work are as follows.

1) We propose AFSIAttention, which dynamically generates frequency-domain weights for different attention heads through head-specific lightweight frequency projection network (HS-LFPN), enabling fine-grained frequency–spatial interaction and facilitating internal integration of frequency and spatial information.
2) We design AFSIFormer, a module that follows a progressive learning strategy by sequentially applying BADAM and LWAM based on AFSIAttention, achieving refined global modeling from coarse to fine under the guidance of frequency-domain information.

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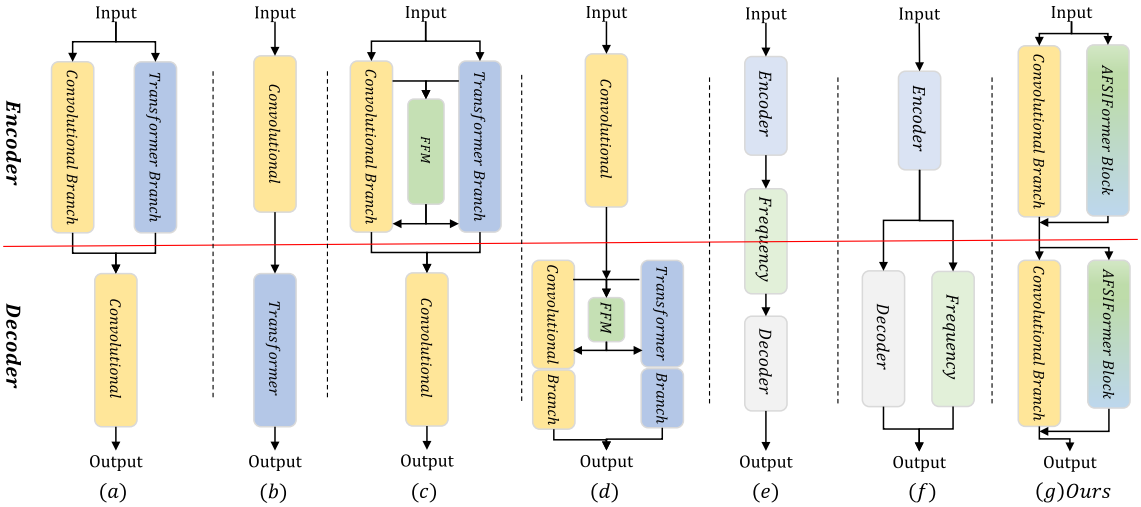


Fig. 2. Compared with the existing mainstream aerial image semantic segmentation frameworks. (a) Backbone-level PC. (b) Backbone-level TC. (c) Backbone-level PC with feature fusion module (FFM). (d) Block-level PC with FFM. (e) Frequency information is only integrated in the deepest layer of the network. (f) Frequency information is only incorporated in the decoder. Our proposed method (g) introduces a block-level residual coupling architecture, which effectively captures global context and frequency-domain information at each stage of the network, thereby enhancing cross-domain information interaction.

3) We construct an efficient semantic segmentation network SynFSNet to ensure continuous propagation of global and frequency information at each stage and mitigate information redundancy caused by the direct fusion of low-level and high-level semantic features.
4) We conduct extensive experiments on three benchmark aerial image semantic segmentation datasets, demonstrating that SynFSNet achieves SOTA performance in accuracy and generalization.

II. Related Work

A. Frequency–Spatial-Domain Information Fusion and Application in Aerial Image Segmentation

Frequency-domain analysis was originally developed in the field of digital signal processing [39] and has been widely applied in classical control theory. In recent years, it has been introduced into computer vision tasks [40], offering a complementary and distinctive theoretical perspective to traditional spatial-domain analysis [41]. As a fundamental tool in frequency-domain analysis, the DCT effectively separates different frequency components of an image, thereby uncovering critical details that are often overlooked in spatial-domain analysis. Specifically, low-frequency components primarily represent large-scale structural features, such as the regular spatial layout of farmlands, while high-frequency components focus on capturing fine object boundaries and texture variations [16], [17]. The DCT exhibits excellent energy compaction properties, effectively concentrating the majority of an image's information into a small number of low-frequency coefficients. This enables the preservation of essential structural information while simultaneously suppressing high-frequency noise. In addition, the strict orthogonality and stable frequency distribution of DCT basis functions provide high controllability and computational stability when applied to fixed windows or local regions, thereby establishing a solid theoretical foundation for subsequent analysis and processing.

Building on these advances, various deep-learning approaches have been proposed to fully exploit frequency-domain information [18], [19], [20], [21], [22], [23]. For instance, ECA-Net [42] introduces a channel-attention mechanism that dynamically adjusts cross-domain feature weights based on the input features. SFFNet [26] employs wavelet transform for fine-grained frequency decomposition, significantly enhancing segmentation accuracy for shadowed grayscale objects. Nevertheless, these methods have notable limitations. First, frequency–spatial integration is often realized through external fusion, where frequency-domain information is introduced only after spatial-domain feature modeling as postprocessing. Second, feature interactions remain simplistic, typically limited to concatenation or addition, which restricts full exploitation of complementary information. Moreover, some methods [22], [23], [26] rely on independent dual-branch architectures and complex fusion modules, substantially increasing parameters and computational cost. Consequently, frequency-domain information is confined to limited stages, leading to its gradual attenuation during propagation. To overcome these issues, the proposed AFSIAttention mechanism embeds frequency information directly within attention computations, enabling fine-grained feature interaction and adaptive weighting. Simultaneously, the block-level residual coupling architecture ensures continuous extraction and propagation of frequency-domain features throughout the network, effectively mitigating its degradation.

B. CNN–Transformer Hybrid Architectures in Aerial Image Segmentation

With rapid advances in deep learning for aerial image semantic segmentation, researchers have increasingly recognized the complementary strengths and limitations of CNNs

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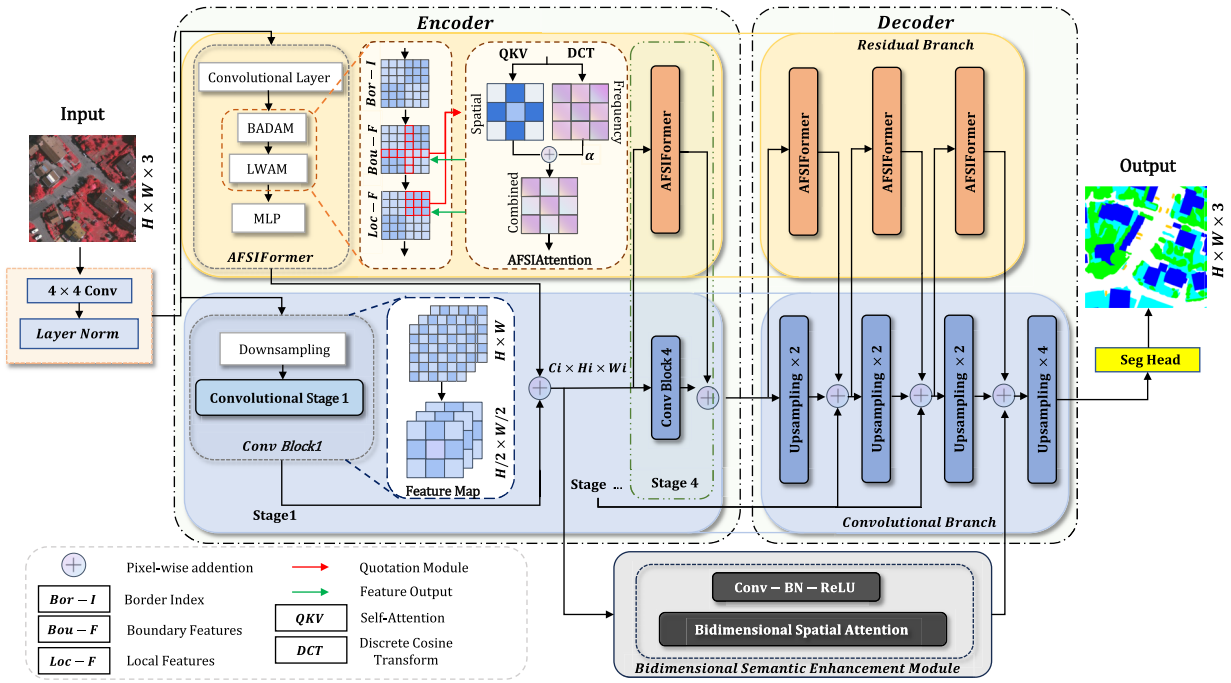


Fig. 3. Overall architecture of SynFSNet. We embed AFSIFormer modules in each stage of the network to enhance the network's perception of global and frequency features and achieve efficient interaction of frequency- and spatial-domain information through AFSIAttention. The BiSEM module bridges the gap between low-level and high-level semantics, promotes feature fusion, and avoids the interference of redundant information on high-level features.

and transformers. CNNs excel at local feature extraction and spatial structure modeling [43], but their fixed receptive fields limit effective global dependency modeling. Conversely, transformers capture global context effectively via self-attention [44] yet often neglect fine-grained spatial details and entail high computational costs. To leverage the advantages of both, hybrid CNN–transformer models [33], [45], [46], [47] have emerged as a leading research direction in aerial image segmentation.

Hybrid architectures for semantic segmentation can be broadly categorized into three organizational forms. The first is the parallel combination (PC) at the backbone level [Fig. 2(a) and (c)], where dual-branch encoders, such as STransFuse [48] and STRD-Net [49], use CNN and transformer branches to extract local and global features, respectively. MCSNet [50] enhances this design by incorporating a dedicated fusion module for more effective integration of local and global information. The second type is the tandem combination (TC) at the backbone level [33], shown in Fig. 2(b), exemplified by networks like TransUNet [51], which sequentially apply CNN-based local feature extraction followed by transformer-based global dependency modeling. The third type is the block-level PC, depicted in Fig. 2(d). UNetFormer [45] exemplifies this by fusing CNN and transformer operations within individual blocks at specific network stages, enabling fine-grained cooperation between local and global feature extraction. A semantic-mask guided semantic segmentation network [38], which integrates transformer and CNN architectures, is pioneeringly developed to effectively enhance the boundary representation of land-cover classes, yielding remarkable improvements in segmentation accuracy.

Although CNN–transformer hybrid architectures [47], [48], [49] have demonstrated remarkable performance, they face several limitations. Dual-encoder frameworks are computationally expensive and resource-intensive, while single-encoder designs struggle to simultaneously capture global and local features, limiting feature representation capacity. Furthermore, the high computational cost of transformers hinders effective propagation of global and local information throughout the network, causing discontinuities in information flow that degrade segmentation accuracy and generalization. To address these issues, we propose a feature-complementary enhanced block-level residual coupling architecture [Fig. 2(g)], which integrates CNN and transformer at every stage of the network. This design effectively mitigates the discontinuity of global information commonly observed in the existing hybrid models.

III. Proposed Method

In this section, we introduce the proposed SynFSNet in detail. Section III-A comprehensively describes the design concept and architecture of SynFSNet. In Section III-B, we elaborate on the operational principles of AFSIFormer, focusing on its network structure and core innovations. In Section III-C, we present the design of the BiSEM module and its core mechanism.

A. Overview of SynFSNet

To efficiently extract global information from images and fully leverage frequency-domain features, we propose the SynFSNet network. The overall architecture is illustrated in Fig. 3. SynFSNet employs a block-level residual coupling

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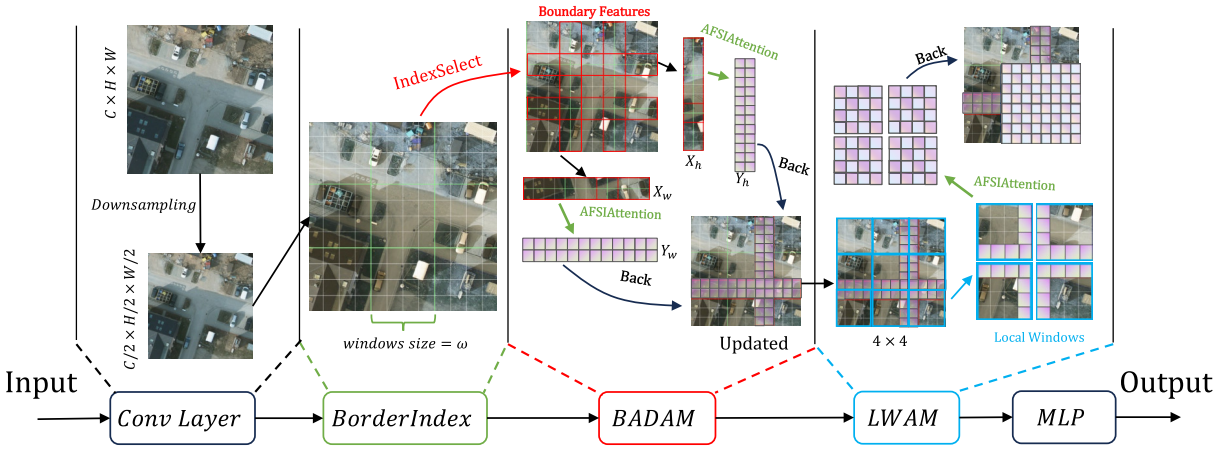


Fig. 4. Illustration of AFSIFormer block. The module adopts a progressive learning strategy. First, boundary indices of window partitioning are extracted through the BorderIndex. Then, in the BADAM module, the extracted boundary features are processed using AFSIAttention to establish long-range contextual relationships. Finally, in the LWAM module, AFSIAttention is applied within local windows to further refine feature information based on global context.

architecture, where an AFSIFormer residual module is integrated at each stage to continuously enhance and efficiently propagate frequency-domain and global semantic information. The AFSIFormer follows a progressive feature learning strategy: it initially captures global structures and edge relationships via the BADAM, followed by modeling of local region features using LWAM. Both mechanisms are based on AFSIAttention to model the contextual information of different windows. Specifically, AFSIAttention generates frequency-domain guidance weights for each attention head and dynamically interacts with spatial-domain attention, enabling fine-grained frequency–spatial information integration at the attention level. The frequency-domain weights can adaptively modulate their contribution based on the semantic characteristics of feature maps at different stages, thereby enhancing the coordination between frequency and spatial representations. To further improve the semantic modeling effect, SynFSNet incorporates the BiSEM module, which leverages a bidirectional spatial attention (SpatAttn) (BiSA) mechanism to selectively enhance key regional features and bridge the representation gap between low-level details and high-level semantics. By enabling fine-grained interaction between frequency-domain and spatial-domain information and collaboratively modeling global and local features, SynFSNet effectively balances segmentation accuracy and computational efficiency. Next, we will provide a detailed explanation of the core modules of SynFSNet.

B. AFSIFormer Block

Aerial image semantic segmentation faces many challenges, including complex and dynamic land cover types, multiscale target distributions, and subtle texture variations. This requires high-precision segmentation models to not only have strong global modeling capabilities but also to accurately perceive detailed features. Global information aids in understanding long-range dependencies between different regions of an image, while frequency-domain information effectively characterizes periodic textures and structural boundaries, thereby enhancing the model's ability to recognize edges and detailed structures. Although the transformer architecture effectively captures global contextual information, its high computational complexity limits its widespread adoption in neural networks. Meanwhile, most existing methods for integrating frequency- and spatial-domain information adopt external integration strategies, lacking fine-grained interactions and dynamic guidance within the mechanism. In addition, complex dual-domain fusion modules often introduce a large number of parameters, making it difficult to fully leverage the complementary strengths of both domains.

To address the above challenges, we propose an efficient AFSIFormer module, as illustrated in Fig. 4. This module follows a progressive feature learning strategy: it first employs BADAM to efficiently capture long-range dependencies across windows while reducing computational complexity. Subsequently, LWAM is applied to extract fine-grained features within local regions, and finally, multilayer perceptron (MLP) is used for semantic enhancement. The core module AFSIAttention is used to model the contextual information of different windows. By introducing the frequency-domain guided weights to dynamically adjust the SpatAttn, the fine modeling of the global context and the deep interaction of the frequency-space features are achieved. Next, we will introduce the overall framework of AFSIFormer and the working principle of the AFSIAttention mechanism in detail.

1) AFSIFormer: While transformers excel at global modeling, their quadratic complexity hampers deployment in aerial image semantic segmentation. Although Swin Transformer [52] reduces complexity through local window partitioning, computational costs remain substantial. We propose AFSIFormer [see Fig. 4], an efficient module that dramatically reduces overhead while preserving strong modeling capabilities. AFSIFormer integrates two core components, BADAM and AFSIAttention, using progressive learning that sequentially applies BADAM followed by LWAM for coarse-to-fine semantic modeling. Specifically, BADAM focuses

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on capturing interwindow contextual information, effectively suppressing redundant feature interference while substantially reducing resource consumption. Meanwhile, AFSIAttention performs fine-grained modeling of contextual dependencies different windows under frequency-domain guidance, enabling efficient frequency-space collaborative modeling. The detailed process of AFSIFormer is as follows.

Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, where C, H, and W denote the number of channels, height, and width, respectively, we first perform spatial downsampling and channel compression via a convolutional layer with a 2×2 kernel and stride 2, followed by batch normalization.

For boundary information modeling, we define a function BorderIndex that selects the first and last positions along a specified axis within each nonoverlapping window of size w and uses them as boundary indices for further processing

$$\text{BorderIndex}\,(h,w) = \left\{ i\cdot w,\ i\cdot w+(w-1) \mid i\in\left[0,\left\lfloor\frac{h}{w}\right\rfloor-1\right]\right\} \tag{1}$$

where h denotes the length of the feature map along a given axis and w is the window size.

We apply this boundary index function along the height and width axes of the downsampled feature map $X_{\text{reduced}} \in \mathbb{R}^{(C/2)\times(H/2)\times(W/2)}$ to extract boundary features

$$X_h = \text{IndexSelect}\left(X_{\text{reduced}},\ \text{dim}=1, \text{BorderIndex}\left(\frac{H}{2},\ w\right)\right) \tag{2}$$

$$X_w = \text{IndexSelect}\left(X_{\text{reduced}},\ \text{dim}=2, \text{BorderIndex}\left(\frac{W}{2},\ w\right)\right). \tag{3}$$

The parameter dim specifies the axis along which the update is performed, with dim = 1 for the height axis and dim = 2 for the width axis.

Subsequently, the AFSIAttention mechanism is applied to the extracted boundary features X_h and X_w for global context modeling, producing enhanced boundary representations Y_h and Y_w along the height and width dimensions, respectively.

Finally, the original boundary features along the height axis are replaced with Y_h, followed by substituting the boundary features along the width axis with Y_w. This process yields the updated feature map X_{updated}, defined as

$$X^{(h)}_{\text{reduced}} = \text{Update}\,(X_{\text{reduced}}, Y_h, \text{dim}=1) \tag{4}$$

$$X_{\text{updated}} = \text{Update}\left(X^{(h)}_{\text{reduced}}, Y_w, \text{dim}=2\right). \tag{5}$$

This strategy reduces model complexity while capturing interwindow contextual information, thereby providing guidance for subsequent LWAM modeling.

After processing the feature maps with BADAM, LWAM is further applied to model intrawindow contextual information, enabling a coarse-to-fine progressive modeling process.

The LWAM divides the direction-enhanced feature map X_{updated} into fixed-size windows (4×4) and applies AFSIAttention within each window. Finally, the feature map is output through an MLP. The calculation process is as follows:

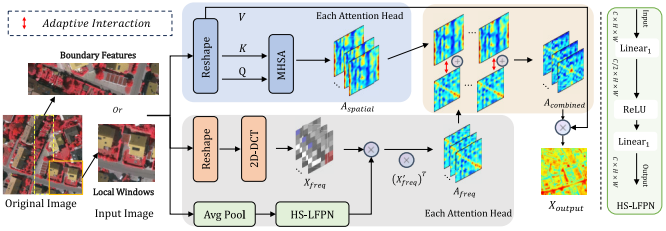


Fig. 5. Illustration of AFSIAttention. Frequency-specific weights for each attention head are generated via HS-LFPN and then dynamically interact with spatial-domain weights through learnable parameters to achieve fine-grained frequency–spatial integration.

$$X_{\text{windows}} = \text{WindowPartition}\left(X_{\text{updated}}, [4,4]\right) \tag{6}$$

$$X_{\text{output}} = \text{MLP}\left(\text{AFSIAttention}\left(X_{\text{windows}}\right)\right) \tag{7}$$

where X_{windows} is a feature map that has been divided into 4×4 windows and X_{output} is the result of applying AFSIAttention in each window.

The output feature map X_{output}, obtained through a progressive feature learning strategy, effectively integrates long-range dependencies with local detail information. This feature map can be seamlessly fused with a CNN-based branch to enhance local detail representation and global semantic understanding.

2) AFSIAttention: To efficiently utilize frequency-domain information, we propose the AFSIAttention mechanism, as illustrated in Fig. 5, achieving internal integration of frequency and spatial domains. This mechanism dynamically generates frequency-guided weights for each attention head via HS-LFPN, enabling fine-grained internal fusion of frequency and spatial information during the attention computation. Without introducing additional complex fusion modules, it maintains a lightweight structure while maximizing the preservation of frequency-domain feature integrity. Specifically, we adopt the computationally efficient DCT, composed of real numbers, to extract frequency features. DCT offers excellent boundary handling capabilities and concentrates the image's primary semantic content into a small number of low-frequency components, allowing the attention mechanism to focus more on semantically critical regions while suppressing redundant details. Leveraging AFSIFormer, we can efficiently introduce frequency-domain guidance in a dynamic and hierarchical adaptive manner while modeling the internal contextual information of the window, so as to more effectively capture texture boundaries and structural information and improve segmentation performance. The process of AFSIAttention is as follows.

Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, we first compute the standard multi-head self-attention (MHSA) to capture spatial dependencies. The feature map is reshaped into a sequence of tokens and processed as follows:

$$X_{\text{flat}} = \text{Reshape}\,(X) \in \mathbb{R}^{N\times C}, \quad N = H\times W \tag{8}$$

$$Q = X_{\text{flat}}W_Q, \quad K = X_{\text{flat}}W_K, \quad V = X_{\text{flat}}W_V \tag{9}$$

$$A_{\text{spatial}} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{10}$$

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where $W_Q, W_K, W_V \in \mathbb{R}^{C\times d}$ are the learnable projection matrices and d_k is the dimension of the key vectors.

To effectively extract DCT features, we first perform channelwise averaging on the input feature map $X \in \mathbb{R}^{B\times N\times C}$ to centralize the frequency domain and remove the direct current component

$$X_{\text{mean}} = \frac{1}{C}\sum_{i=1}^{C} X[:,:,i] \in \mathbb{R}^{B\times N} \tag{11}$$

We reshape the feature map from $\mathbb{R}^{B\times N\times C}$ to $\mathbb{R}^{B\times N}$ by computing the channel mean. We then reshape the feature map into a 2-D layout and apply a 2-D DCT to extract frequency-domain features X_{freq}

$$X_{2D} = \text{Reshape}\left(X_{\text{mean}}\right) \in \mathbb{R}^{B\times H\times W} \tag{12}$$

$$X_{\text{freq}} = \text{DCT}_{2D}\left(X_{2D}\right) \in \mathbb{R}^{B\times H\times W}. \tag{13}$$

We propose an HS-LFPN that generates customized frequency-domain weights for different attention heads and adaptively adjusts their corresponding frequency responses. This design enhances the adaptability of frequency-guided modeling while preserving the integrity of frequency-domain information. The computational process is described as follows:

$$W_{\text{head}} = \text{HS-LFPN}\left(X_{\text{avg}}\right) \tag{14}$$

$$\text{HS-LFPN}\left(X_{\text{avg}}\right) = \text{Linear}_2\left(\text{ReLU}\left(\text{Linear}_1\left(X_{\text{avg}}\right)\right)\right) \tag{15}$$

$$X'_{\text{freq}} = X_{\text{freq}} \otimes W_{\text{head}} \tag{16}$$

where X_{avg} denotes the feature obtained via average pooling, $W_{\text{head}} \in \mathbb{R}^{B\times h}$ represents the head-specific frequency weights, and $\otimes$ denotes the elementwise weighting. Based on the weighted frequency-domain features X'_{freq}, we construct the frequency-domain attention map A_{freq} as follows:

$$A_{\text{freq}} = X'_{\text{freq}} \cdot \left(X'_{\text{freq}}\right)^T \tag{17}$$

where A_{freq} encodes the pairwise correlations between tokens in the frequency domain through transposed matrix multiplication. The frequency-domain attention map effectively captures boundary information and accurately characterizes pixel-level similarity in the frequency domain by customizing frequency-aware representations for each attention head.

Finally, we introduce a learnable adaptive weight α to facilitate deep interaction and complementary integration between frequency-domain attention map and standard SpatAttn map

$$A_{\text{combined}} = (1-\sigma(\alpha))\cdot A_{\text{spatial}} + \sigma(\alpha)\cdot A_{\text{freq}} \tag{18}$$

where $\sigma(\cdot)$ denotes the sigmoid function, mapping α to the range $(0,1)$ to ensure a balanced contribution from both attention mechanisms. A_{spatial} represents the conventional dot-product SpatAttn, while A_{freq} captures the frequency-domain dependencies. This adaptive interaction enables the model to dynamically adjust the relative contributions of the two attention branches based on information from different levels, thereby enhancing its ability to capture complex structural patterns.

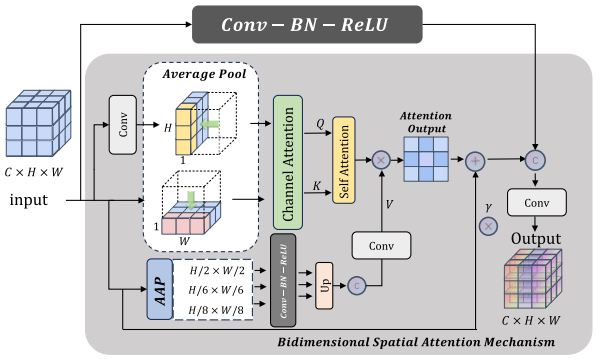


Fig. 6. Illustration of BiSEM. Employs a bidimensional attention mechanism to effectively bridge the gap between low-level and high-level semantics.

The combined attention map is subsequently used to weight the values V and perform feature fusion, yielding a representation that deeply integrates frequency-domain and spatial-domain information

$$X_{\text{output}} = A_{\text{combined}} \cdot V. \tag{19}$$

C. BiSEM Block

In aerial image segmentation, features at different levels encode distinct and complementary information. Low-level features often capture fine details, whereas high-level features represent more abstract semantic concepts. Fusing these features enables the model to simultaneously preserve detailed boundaries and understand global context, thereby improving segmentation accuracy. However, direct fusion via residual connections often introduces redundant low-level details, which interfere with high-level semantic representations and ultimately degrade segmentation performance.

To address this issue, we propose BiSEM (illustrated in Fig. 6), which replaces traditional residual connections with an efficient feature fusion method. BiSEM consists of two branches and three core mechanisms. The two branches are the basic transformation branch and the BiSA mechanism branch, while the three core mechanisms include SpatAttn decomposition, a learnable fusion gate γ, and a multiscale context-aware component V. These refined structures preserve the most valuable fine-grained information. After processing by BiSEM, low-level features can be more effectively integrated with high-level representations, thereby improving overall segmentation performance.

The BiSEM architecture, shown in Fig. 6, comprises two parallel branches: a basic transformation branch and a BiSA mechanism branch. Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, the first step is a 1×1 convolution for feature transformation

$$X_1 = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times 1}(X)\right)\right). \tag{20}$$

This transformation enhances global feature retention while maintaining computational efficiency. The basic transformation branch and the BiSA branch work together to improve feature expressiveness.

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The core of BiSEM is the BiSA mechanism, where we innovatively decompose SpatAttn calculation into two dimensions, H and W, to obtain Q and K

$$X_Q = \text{Conv}_{3\times3}(X)$$
$$Q = \text{ChannelAttention}\left(\text{AveragePool}\left(X_Q, [H, 1]\right)\right) \quad (21)$$
$$K = \text{ChannelAttention}\left(\text{AveragePool}\left(X, [1, W]\right)\right). \quad (22)$$

To capture contextual dependencies at multiple receptive field sizes, we employ adaptive average pooling (AAP) with kernel sizes $[H/2, W/2]$, $[H/6, W/6]$, and $[H/8, W/8]$. Each pooled feature is processed through a shared transformation pipeline consisting of a 1×1 convolution, batch normalization, and ReLU activation. The resulting feature maps are then upsampled via bilinear interpolation and concatenated along the channel dimension. Finally, a 1×1 convolution is applied to generate the value representation V.

We calculate the attention scores and normalize them

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \quad (23)$$

where d is the square root of the feature dimension, used to scale the attention scores. Then, we weight the value features by the attention weights and fuse them adaptively using a learnable parameter γ

$$X_2 = \gamma \cdot X + A \cdot V. \quad (24)$$

Finally, BiSEM integrates the transformed features with the adaptive attention features through feature dimension concatenation

$$Y = \text{Conv}_{1\times1}\left(\text{Concat}\left[X_1, X_2\right]\right). \quad (25)$$

This strategy enables the model to preserve the integrity of the original features while integrating rich spatial contextual information, effectively bridging the gap between low-level and high-level semantics.

IV. Datasets and Experimental Settings

A. Datasets

We evaluate our method on three high-resolution aerial datasets: ISPRS Vaihingen [53], ISPRS Potsdam [53], and LoveDA [54]. These datasets have different scene layouts and annotation protocols, posing different challenges for semantic segmentation. Fig. 7 compares these three datasets.

1) ISPRS Vaihingen: The Vaihingen dataset, released by the International Society for Photogrammetry and Remote Sensing, consists of 33 true orthophoto (TOP) images with an average resolution of 2500 × 2000 pixels and a ground sampling distance (GSD) of 9 cm, covering urban areas in Germany. In our experiments, we only utilize the RGB images, excluding the provided digital surface model (DSM) data. The dataset is annotated with six semantic categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. Following the standard protocol [45], 16 images with IDs 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 30, 32, 34, and 37 are used for training, while the remaining 17 images are reserved for testing.

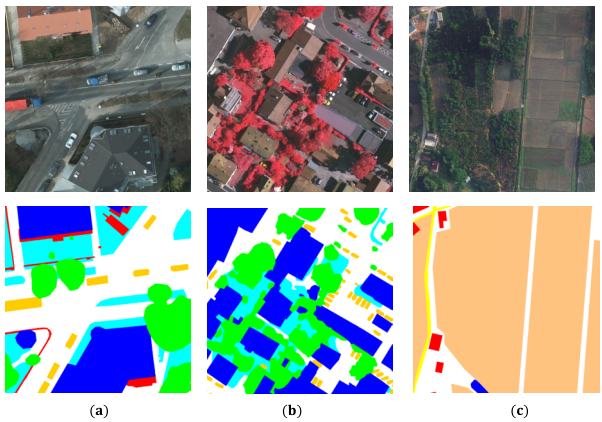


Fig. 7. Comparison of features of different aerial semantic segmentation datasets. (a) ISPRS Potsdam. (b) ISPRS Vaihingen. (c) LoveDA.

2) ISPRS Potsdam: The Potsdam dataset is also widely used for semantic segmentation in aerial and shares a similar structure with the Vaihingen dataset. It consists of 38 high-quality image tiles with a resolution of 6000 × 6000 pixels and a GSD of 5 cm. In this study, we use only the RGB bands and exclude the image 7_10 due to annotation errors. The semantic class definitions are consistent with those of the Vaihingen dataset. Following the common protocol [45], 23 images are used for training, while the remaining 14 images are used for testing.

3) LoveDA: LoveDA is a large-scale aerial semantic segmentation dataset covering both urban and rural areas. It consists of 5987 images with a resolution of 1024 × 1024 pixels and a GSD of 0.3 m. The dataset spans three cities in China—Nanjing, Changzhou, and Wuhan—and includes annotations for seven land cover categories: buildings, roads, water, vegetation, vehicles, clutter, and background. Following the standard split, the dataset contains 2522 training images, 1669 validation images, and 1796 test images. With significant scene diversity and domain distribution shifts, LoveDA serves as a challenging benchmark for evaluating the generalization ability of segmentation models.

B. Experimental Details

1) Dataset Partitioning and Preprocessing Methods: To ensure the fairness of comparative experiments, we adopt a unified data preprocessing strategy consistent with previous studies [45]. Specifically, for the ISPRS Vaihingen and Potsdam datasets, large images are cropped into patches of size 1024 × 1024 for training and testing. In the Vaihingen dataset, 16 images are used for training, and the remaining 17 are reserved for testing. For the Potsdam dataset, 23 image tiles are selected for training and 14 for testing. For the LoveDA dataset, since the original images are already of size 1024 × 1024, no cropping is required, and the data are directly divided into training, validation, and test subsets for experimentation.

2) Detailed Experimental Setup: All experiments are conducted using the PyTorch framework on a workstation equipped with an NVIDIA RTX 4090 GPU (24 GB). The

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TABLE I
COMPARISON WITH SOTA METHODS ON THE ISPRS VAIHINGEN DATASET

Structures	Methods	F1(%)					Evaluation metrics		
		Impervious surface	Building	Low vegetation	Tree	Car	mIoU(%)	OA(%)	mF1
CNN-Based Models	FCN8s [55]	95.48	92.12	82.03	88.80	79.96	78.55	89.22	87.68
	UNet [56]	90.50	93.30	79.60	87.50	76.40	75.50	89.20	85.50
	DeepLabv3+ [57]	92.27	95.40	83.56	89.22	85.36	80.73	90.27	89.16
	DANet [58]	91.38	94.10	83.09	89.02	76.79	77.32	89.20	86.88
	PSPNet [59]	91.17	94.64	82.92	88.99	73.41	76.51	89.49	86.23
	BiSeNet [60]	89.10	91.30	80.91	86.91	73.07	75.81	87.10	84.30
	FastSCNN [61]	89.80	91.97	80.89	87.86	67.31	72.49	87.55	83.40
CNN-Trans Hybrid Models	SwinUperNet [52]	90.26	94.07	81.13	87.04	81.58	77.05	88.21	86.82
	Segmenter [62]	89.80	93.00	81.20	88.90	67.60	73.60	88.10	84.10
	ST-UNet [31]	91.40	94.10	83.20	89.10	82.60	77.20	88.60	88.10
	BANet [63]	92.20	95.20	83.80	89.90	86.80	81.40	90.50	89.60
	MA-Net [10]	93.02	95.47	84.64	89.98	88.95	82.71	90.96	90.41
	A^2-FPN [64]	96.59	95.30	83.27	89.37	86.93	82.68	90.78	90.29
	UNetFormer [45]	96.80	95.49	83.61	89.36	87.62	83.14	90.99	90.57
	CMTFNet [33]	96.74	95.70	84.23	89.76	88.31	83.74	91.15	90.95
	DC-Swin [47]	93.46	96.00	85.30	90.03	84.88	82.01	91.91	89.94
	TransUNet [51]	92.80	95.40	84.40	89.90	85.30	81.30	90.70	89.50
	ESDINet [65]	92.74	95.54	84.46	90.03	87.16	82.03	90.88	89.99
	CGGLNet [66]	93.34	96.10	85.30	90.33	90.50	83.89	91.47	91.12
	FT-UNetformer [45]	93.50	96.00	85.60	**90.80**	90.40	84.10	91.60	91,30
	SLCNet [67]	92.74	95.70	84.80	90.30	89.08	82.90	90.81	90.53
	MCSNet [50]	93.52	95.85	85.83	91.12	90.48	84.26	91.97	91.36
	SparseFormer [68]	96.72	95.65	84.42	89.93	88.85	84.12	91.21	91.11
Freq-Enhanced Models	DualM [69]	93.20	95.50	84.60	89.80	88.90	82.70	91.40	90.40
	SFFNet [26]	96.86	95.57	84.79	90.13	89.27	84.38	91.36	91.32
	XNet [70]	96.81	95.63	84.96	90.32	89.29	84.43	91.52	91.40
	AFENet [71]	96.90	95.72	85.07	90.64	89.37	84.55	91.67	91.54
Ours	SynFSNet	**97.20**	**96.40**	**85.31**	90.58	**91.27**	**85.74**	**93.85**	**92.15**

training pipeline follows the default configuration of the UNetFormer [45] implementation provided by the GeoSeg framework. Specifically, we employ the AdamW optimizer with an initial learning rate of 6×10^{-4} and a cosine annealing learning rate schedule with linear warm-up. The batch size is set to 4. To enhance generalization, standard data augmentation strategies, such as random horizontal flipping and random scaling, are applied. The number of training epochs is set to 105 for the ISPRS Vaihingen and Potsdam datasets and 30 for the LoveDA dataset, in accordance with the GeoSeg default settings. The CNN backbone is initialized with pretrained weights on the ImageNet-1K dataset, while the remaining modules are initialized randomly.

C. Evaluation Metrics

To quantitatively evaluate the performance of our model, we adopt three widely used metrics: overall accuracy (OA), mean intersection over union (mIoU), and $F1$-score (F1). The specific formulations are as follows:

$$\mathrm{OA} = \frac{\sum_i \mathrm{TP}_i}{\sum_i (\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i)} \tag{26}$$

$$\mathrm{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i}. \tag{27}$$

The precision and recall for class i are defined as

$$P_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i}, \quad R_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FN}_i}. \tag{28}$$

Then, the $F1$-score for class i is computed as

$$\mathrm{F1}_i = \frac{2 \cdot P_i \cdot R_i}{P_i + R_i} \tag{29}$$

and the overall $F1$-score is given by

$$\mathrm{F1} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{F1}_i \tag{30}$$

where TP_i, FP_i, and FN_i denote the number of true positives, false positives, and false negatives for class i, respectively, P_i and R_i represent the precision and recall, and N is the total number of classes.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Comparison With SOTA Methods

To validate the effectiveness and generalization of our proposed model, we conduct extensive comparisons on three widely used aerial datasets against a range of baseline and state-of-the-art (SOTA) methods for semantic segmentation. The compared methods include CNN-based approaches [55], [56], [57], [58], [59], [60], [61], hybrid CNN–transformer architectures [10], [31], [33], [45], [47], [50], [51], [52], [62], [63], [64], [65], [66], [67], [68], [72], [73], [74], [75], [76], as well as enhanced models incorporating frequency-domain information (Freq-Enhanced) [26], [69], [70], [71].

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TABLE II
COMPARISON WITH SOTA METHODS ON THE ISPRS POTSDAM DATASET

Structures	Methods	F1(%)					Evaluation metrics		
		Impervious surface	Building	Low vegetation	Tree	Car	mIoU(%)	OA(%)	mF1
CNN-Based Models	FCN8s [55]	90.95	92.84	85.33	86.25	92.80	81.37	88.34	89.64
	UNet [56]	92.66	94.89	86.17	87.71	95.36	84.36	89.94	91.35
	DeepLabv3+ [57]	93.15	95.43	86.45	87.78	95.21	84.42	90.07	91.40
	DANet [58]	91.94	96.35	86.00	87.67	86.01	81.40	89.73	89.60
	PSPNet [59]	90.79	95.15	86.57	87.32	91.05	82.25	89.31	90.18
	BiSeNet [60]	91.07	95.12	85.81	86.63	91.37	81.99	89.06	90.00
	FastSCNN [61]	90.45	94.48	85.35	85.27	91.84	81.15	88.29	89.48
CNN-Trans Hybrid Models	SwinUperNet [52]	91.60	96.04	86.09	87.00	91.70	82.82	89.43	90.49
	Segmenter [62]	91.50	95.30	85.40	85.00	88.50	80.70	88.70	89.20
	ST-UNet [31]	92.77	96.90	86.45	88.11	95.80	85.46	90.53	92.01
	BANet [63]	93.30	96.70	87.40	89.10	96.00	86.30	91.00	92.50
	MA-Net [10]	93.24	95.64	86.83	88.15	95.83	85.31	90.82	91.94
	A^2-FPN [64]	93.43	96.12	86.89	88.38	95.17	85.53	90.83	92.00
	UNetFormer [45]	93.78	96.20	87.52	88.44	96.17	86.14	91.21	92.42
	CMTFNet [33]	93.28	97.07	87.08	88.42	95.93	86.05	90.92	92.36
	DC-Swin [47]	93.07	96.68	87.42	88.63	95.51	85.86	90.84	92.26
	TransUNet [51]	92.60	96.00	86.80	87.90	95.30	84.90	90.10	91.70
	ESDINet [65]	92.68	96.28	87.29	88.10	95.42	85.32	90.53	91.96
	CGGLNet [66]	93.84	**97.44**	87.66	89.18	96.69	87.11	91.58	92.96
	FT-UNetformer [45]	93.70	97.10	88.30	**89.80**	96.50	87.30	91.60	93.10
	SLCNet [67]	93.21	96.99	87.20	88.99	95.60	86.10	90.94	92.40
	MCSNet [50]	93.59	97.09	88.15	89.63	95.98	86.92	91.40	92.89
	SparseFormer [68]	93.56	96.00	87.55	89.20	96.15	86.83	91.56	92.49
Freq-Enhanced Models	DualM [69]	93.20	97.00	86.90	88.80	95.90	86.00	90.80	92.40
	SFFNet [26]	93.40	96.62	87.73	89.20	96.33	86.93	91.72	92.66
	XNet [70]	93.25	96.57	87.83	89.43	96.43	87.14	91.83	92.70
	AFENet [71]	**94.57**	96.86	88.17	89.78	96.83	87.50	**92.03**	93.24
Ours	SynFSNet	93.92	97.35	**88.58**	89.64	**96.95**	**87.63**	91.87	**93.29**

1) Results on the ISPRS Vaihingen Dataset: The evaluation results of the existing SOTA methods on the ISPRS Vaihingen test set are shown in Table I, and the proposed SynFSNet achieves the best performance among all compared methods on the ISPRS Vaihingen dataset, with an mIoU of 85.74%, OA of 93.85%, and average $F1$-score of 92.15%. SynFSNet outperforms mainstream approaches across four evaluated semantic categories, achieving $F1$-scores of 97.20% for impervious surfaces, 96.40% for buildings, 85.31% for low vegetation, 90.58% for trees, and 91.27% for cars. Compared with the SOTA CNN–transformer hybrid model MCSNet, SynFSNet improves mIoU and OA by 1.48% and 1.88%, respectively. It also surpasses the frequency-domain-enhanced model AFENet, with improvements of 1.19% in mIoU and 0.61% in average $F1$-score. Our model achieves outstanding performance in segmenting small objects, such as car, as well as in categories with ambiguous boundaries, such as tree and building, highlighting the efficiency of the proposed AFSIFormer module in global context modeling and boundary extraction.

2) Results on the ISPRS Potsdam Dataset: Our experimental results for the ISPRS Potsdam test set are shown in Table II. SynFSNet also demonstrates outstanding performance on the ISPRS Potsdam dataset, achieving the best results among all compared methods, with an mIoU of 87.63% and an mF1 of 93.29%. Across the five semantic classes, SynFSNet achieves either the best or near-best $F1$-scores: 93.92% for impervious surfaces, 97.35% for buildings, 88.58% for low vegetation, 89.64% for trees, and 96.95% for cars. Compared with the strong baseline UNetFormer, which is a CNN–transformer hybrid architecture, SynFSNet improves mIoU and mF1 by 1.49% and 0.87%, respectively. It also outperforms the frequency-domain enhanced model SFFNet, with improvements of 0.70% in mIoU and 0.63% in $F1$-score. Notably, SynFSNet shows significant advantages in key categories, such as buildings, low vegetation, and cars.

3) Results on the LoveDA Dataset: Our experimental results for the LoveDA test set are shown in Table III. SynFSNet achieves the best overall segmentation performance on the LoveDA dataset, with an mIoU of 53.42%, significantly outperforming the existing SOTA methods. In particular, SynFSNet obtains the highest mIoU scores in four categories: 61.37% for buildings, 80.53% for water, 22.60% for barren, and 49.00% for forest. The performance on the remaining classes is also competitive, demonstrating strong generalization ability across diverse urban and rural scenes. Compared to the current leading method LOGCAN++, SynFSNet achieves a 0.07% improvement in overall mIoU. Its advantage is especially pronounced in small-sample or boundary-ambiguous classes, such as barren and forest, highlighting its effectiveness in capturing fine-grained land cover features.

B. Cross-Domain Evaluation on LoveDA Dataset

To further evaluate the cross-domain generalization ability of our model, we conduct domain adaptation experiments

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TABLE III
EXPERIMENTAL RESULTS OF DIFFERENT METHODS ON THE LOVEDA DATASET

Method	Background	Building	Road	Water	Barren	Forest	Agricultural	mIoU (%)
FCN8s [55]	42.91	50.33	46.82	76.38	17.76	44.39	57.71	48.04
UNet [56]	43.38	53.53	53.52	76.03	17.75	44.74	58.33	49.61
DeepLabv3+ [57]	42.97	50.88	52.02	74.36	10.40	44.21	58.53	47.62
PSPNet [59]	44.40	52.13	53.52	76.50	9.73	44.70	57.85	48.31
BANet [63]	43.70	51.50	51.10	76.90	16.60	44.90	62.50	49.60
MA-Net [10]	38.70	51.70	42.60	72.60	15.30	42.10	57.70	45.70
TransUNet [51]	43.00	56.10	53.70	78.00	9.30	44.90	56.90	48.90
Segmenter [62]	38.00	50.70	48.70	77.40	11.30	43.50	58.20	47.10
DCSwin [47]	41.30	54.50	56.20	78.10	14.50	47.20	62.40	50.60
SwinUperNet [52]	43.30	54.30	54.30	78.70	14.90	45.30	59.60	50.00
UNetFormer [45]	44.70	58.80	54.90	79.60	20.10	46.00	62.40	52.40
EfficientViT [72]	42.90	51.00	52.80	75.70	4.30	42.00	61.20	47.10
DDP [73]	46.20	57.20	**58.20**	80.30	14.90	46.50	64.30	52.50
RSSFormer [74]	44.03	57.23	55.35	79.37	10.91	45.22	62.04	50.59
BiFormer [76]	43.60	55.30	55.90	79.50	16.90	45.40	61.50	51.20
ESDINet [65]	41.57	53.77	54.84	78.71	19.49	44.17	57.95	50.07
LOGCAN++ [75]	**47.37**	58.38	56.46	80.05	18.44	47.91	**64.80**	53.35
SynFSNet	45.36	**61.37**	56.61	**80.53**	**22.60**	**49.00**	58.46	**53.42**

TABLE IV
CROSS-DOMAIN EVALUATION ON LOVEDA DATASET

Method	Urban → Rural		Rural → Urban	
	mIoU (%)	mF1-score (%)	mIoU (%)	mF1-score (%)
DeepLabv3+ [57]	35.97	50.12	52.15	68.07
MA-Net [10]	35.97	51.44	50.70	66.74
UNetformer [45]	35.32	49.68	50.73	67.07
A^2-FPN [64]	33.55	47.03	51.76	67.67
SynFSNet (Ours)	**44.40**	**59.59**	**56.24**	**71.50**

on the LoveDA dataset, where the model is trained on one domain (urban or rural) and tested on the other. As shown in Table IV, all results are reported on the validation set. Among the compared methods, SynFSNet consistently achieves the best performance in both domain adaptation directions. Specifically, under the urban → rural setting, SynFSNet attains an mIoU of 44.40% and an mF1 score of 59.59%, significantly outperforming the second-best model. Similarly, in the rural → urban scenario, it achieves an mIoU of 56.24% and an mF1 score of 71.50%, demonstrating strong transferability across heterogeneous domains.

C. Robustness Evaluation

To evaluate the robustness of our model under typical input perturbations, we conduct comprehensive experiments on the Vaihingen dataset using three commonly observed degradation types: additive Gaussian noise (variance = 0.01), Gaussian blur (kernel size = 3), and random occlusion (with 5% occluded area). As shown in Table V and Fig. 8, traditional CNN-based methods, such as DeepLabv3+ and MA-Net, suffer significant performance degradation under noise and occlusion, with DeepLabv3+ exhibiting a 4.3% drop under noise and a 10.45% drop under occlusion. Transformer-based methods, including UNetformer and A^2-FPN, achieve relatively better performance on clean inputs but show notable declines under noise perturbations. In particular, A^2-FPN drops sharply from 82.68% to 57.93% when exposed to noise. In contrast, our

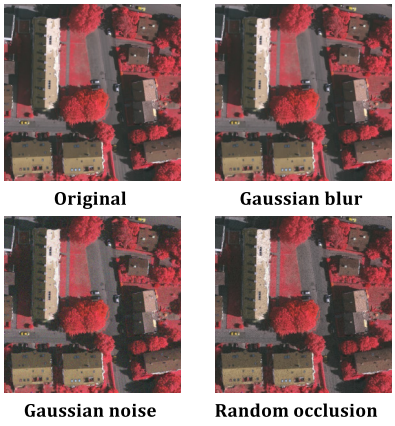


Fig. 8. Visual comparison of original and corrupted inputs under different noise types.

TABLE V
ROBUSTNESS EVALUATION UNDER TYPICAL INPUT PERTURBATIONS

Method	Clean	Noise (0.01)	Blur (k=3)	Occlusion (5%)
DeepLabv3+ [57]	80.73%	76.43%	79.17%	70.28%
MA-Net [10]	82.71%	77.34%	80.39%	72.19%
UNetformer [45]	83.14%	67.49%	82.31%	72.75%
A^2-FPN [64]	82.68%	57.93%	81.40%	73.84%
SynFSNet (Ours)	**85.74%**	**84.80%**	**84.90%**	**73.84%**

proposed SynFSNet demonstrates strong robustness, with only a 0.94% performance drop under noise and minimal impact from blur and occlusion.

D. Efficiency Analysis

To comprehensively evaluate the efficiency and practicality of the proposed SynFSNet, we compare its model complexity and computational cost with several representative methods on the ISPRS Vaihingen dataset under identical experimental settings, as summarized in Table VI. Model complexity is

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TABLE VI
COMPARISON OF MODEL COMPLEXITY AND SEGMENTATION PERFORMANCE

Method	Params	FLOPs	MACs	Model Size	mIoU (%)
FCN8s [55]	**30.03M**	320.87G	160G	**114.6MB**	78.55
PSPNet [59]	46.6M	183.37G	92G	177.6MB	76.51
DeepLabv3+ [57]	59.3M	181.20G	91G	226.2MB	80.73
DANet [58]	66.4M	217.58G	109G	253.3MB	77.32
ST-UNet [31]	161.0M	-	-	613.8MB	77.20
TransUNet [51]	105.9M	168.90G	84G	404.2MB	81.30
MA-Net [10]	35.68M	310.47G	155G	136.1MB	82.71
DC-Swin [47]	66.95M	276.32G	138G	255.6MB	83.22
CMTFNet [33]	30.07M	130.0G	65G	114.7MB	83.74
FT-UNetformer [45]	96.0M	128.40G	64G	366.3MB	84.10
CGGLNet [66]	34.83M	605.0G	302G	132.9MB	83.89
LOGCAN++ [75]	31.05M	**51.26G**	**26G**	118.4MB	81.93
SynFSNet	68.43M	73.56G	37G	260.6MB	**85.74**

measured by parameters (Params) and model size, while computational cost is quantified by floating-point operations (FLOPs) and multiply–accumulate operations (MACs) per forward pass.

SynFSNet achieves the highest segmentation accuracy of 85.74% mIoU while maintaining moderate parameter size (68.43M) and reasonable model size (260.6 MB). More importantly, it exhibits exceptionally low computational cost, requiring only 73.56 GFLOPs and 37 GMACs—substantially lower than most mainstream methods. For instance, FCN8s and MA-Net consume 320.87G/160G and 310.47G/155G FLOPs/MACs, respectively, yet yield much lower mIoUs of 78.55% and 82.71%. High-performing models, such as dc-Swin (83.22%) and FT-UNetformer (84.10%), also incur higher computational burdens with 276.32/138 G and 128.40G/64G FLOPs/MACs. Although lightweight models, such as CMTFNet (30.07 M, 114.7 MB) and LOGCAN++ (31.05 M, 118.4 MB), show strong compactness, their accuracy (83.74%, 81.93%) remains inferior to SynFSNet. This result highlights the lightweight nature of the proposed block-level residual coupling architecture.

E. Ablation Experiment

To comprehensively evaluate the effectiveness and contribution of each module, we designed a series of ablation experiments on the ISPRS Vaihingen dataset, a commonly used benchmark for semantic segmentation of aerial images.

1) Effectiveness of SynFSNet Core Modules: To comprehensively evaluate the effectiveness of each core module of SynFSNet, we conducted ablation experiments on the ISPRS Vaihingen dataset. The results are shown in Table VII and Fig. 9. Using the classic UNet architecture as the baseline, we analyzed the individual and combined effects of the BADAM and LWAM in the AFSIFormer, as well as the BiSEM. The experiments demonstrate that the introduction of AFSIFormer alone effectively enhances the model's ability to capture global context and boundary details, leading to a significant improvement in segmentation performance. The BiSEM module enhances the ability to express detailed features through cross-layer semantic fusion, which also brings performance improvements. When the two modules are used together, the model achieves further improvement in accuracy,

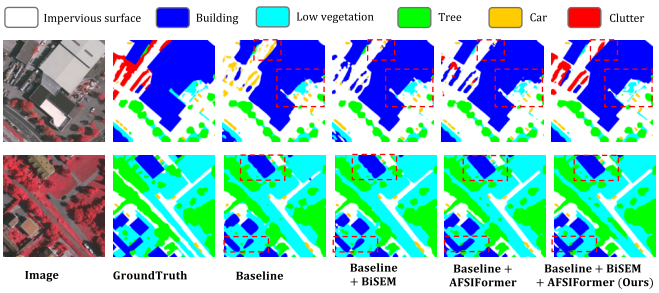


Fig. 9. Ablation experiment visualization of core modules in SynFSNet.

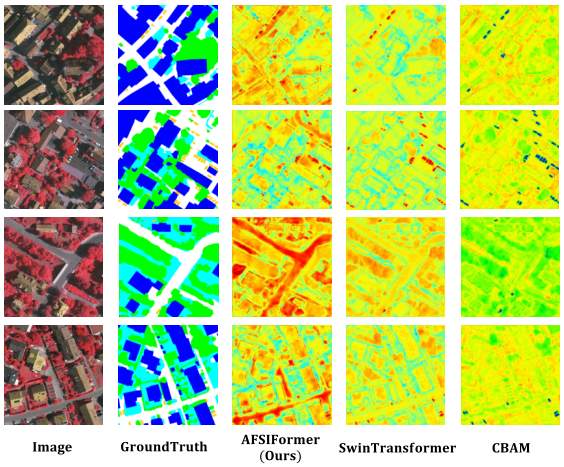


Fig. 10. Comparison of heatmaps produced by different transformer blocks in the final layer of the network.

indicating that the two modules have good complementarity and synergy in semantic enhancement and context modeling. We also evaluated the BADAM and LWAM modules separately. The results showed that the performance improvement brought by introducing BADAM or LWAM alone is relatively limited, but the synergy between the two can give full play to their respective advantages, achieving a better balance between boundary perception and local context modeling, and significantly enhance the overall feature expression capability. In terms of model complexity, AFSIFormer and BiSEM only add a small amount of parameters and FLOPs but achieve a significant improvement in accuracy, showing a good performance-efficiency balance. Fig. 9 presents the typical visualization results. The introduction of AFSIFormer into the baseline model significantly improves the segmentation performance for small targets with blurred boundaries, such as buildings and vehicles. The addition of BiSEM further enhances the model's capability to extract fine details and understand global information. When both modules are combined, the model achieves more precise segmentation at both boundary and detail levels, demonstrating the effectiveness of the proposed modules.

2) Efficiency Analysis of AFSIFormer Block: To verify the efficiency of the AFSIFormer module we designed, we replace AFSIFormer with other attention modules for comparative analysis, as shown in Table VIII. The results indicate that although classical attention modules (such as CCA [77] and

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TABLE VII
ABLATION STUDY OF KEY COMPONENTS IN SYNFSNET

Method	AFSIFormer		BiSEM	Performance			
	BADAM	LWAM		mIoU(%)	mF1(%)	Params(M)	FLOPs(G)
Baseline+	✗	✗	✗	83.07	90.52	58.42	68.88
	✓	✓	✗	84.72	91.55	68.21	71.05
	✗	✗	✓	83.93	91.05	58.65	71.39
	✓	✗	✓	84.29	91.28	68.43	73.27
	✗	✓	✓	84.36	91.32	68.43	73.44
SynFSNet	✓	✓	✓	**85.74**	**92.15**	68.43	73.56

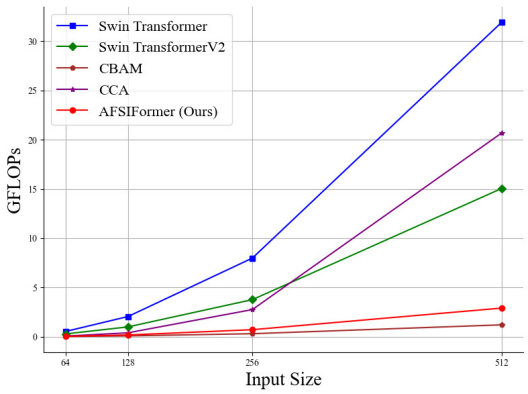


Fig. 11. Comparison of the computational complexity between AFSIFormer and recent advanced transformer blocks under various input feature map resolutions.

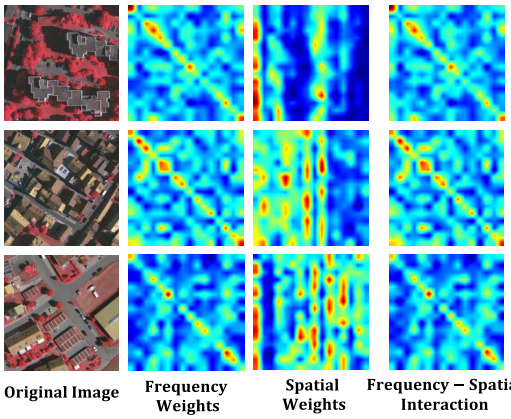


Fig. 12. Visualization of frequency, spatial, and interaction attention maps for different inputs.

TABLE VIII
ABLATION STUDY ON THE EFFICIENCY OF AFSIFORMER

Method Variant	mIoU (%)	mF1 (%)	FLOPs(G)
Baseline	83.93	91.05	71.39
+Swin Transformer [52]	84.42	91.37	102.31
+CCA [77]	84.60	91.47	73.91
+CBAM [78]	84.68	91.53	**72.49**
+Swin TransformerV2 [79]	84.77	91.57	87.81
+AFSIFormer (Ours)	**85.74**	**92.15**	73.56

CBAM [78]) and mainstream transformer architectures (such as the Swin Transformer [52] and its V2 variant [79]) all improve model performance to some extent, there are significant differences in computational overhead. Specifically, while the Swin Transformer achieved a 0.49% increase in mIoU, it incurred a 43% increase in FLOPs, reflecting low efficiency. In contrast, benefiting from the design of BADAM, AFSIFormer achieves a notable performance improvement with only a slight increase in computational cost (from 71.39 to 73.56 G, approximately 3%). It raises the mIoU to 85.74% and achieves an mF1 score of 92.15%, outperforming all other compared modules in terms of key performance metrics. Overall, our AFSIFormer not only surpasses the Swin Transformer series in terms of accuracy but is also superior to lightweight attention mechanisms, such as CBAM, in terms of computational efficiency, achieving a better balance between accuracy and efficiency. Fig. 10 shows the last layer of heatmaps generated after replacing Swin Transformer and CBAM into the network, respectively. The results show that, in comparison, our AFSIFormer pays more attention to the boundary information and detail features in the image, reflecting a stronger feature perception ability. Fig. 11 illustrates the computational complexity comparison between our proposed AFSIFormer and other mainstream transformer blocks under varying input resolutions. It can be observed that as the input resolution increases from 64 × 64 to 512 × 512, the computational complexity of AFSIFormer increases by only about 2.86G, demonstrating its superior efficiency and scalability for high-resolution feature maps.

3) Effectiveness of Adaptive Frequency–Spatial Interaction in AFSIAttention: To validate the effectiveness of our proposed adaptive frequency–spatial interactive mechanism in aerial semantic segmentation tasks, we conducted comparative experiments based on a baseline model without attention modules. We progressively introduced the proposed frequency attention (FreqAttn), SpatAttn, and different fusion strategies, as shown in Table IX. The experimental results show that when FreqAttn is introduced alone, the mIoU slightly decreases (from 83.93% to 83.75%), indicating that although frequency-domain information has the potential for global semantic modeling, its expressive capability is limited without the collaboration of spatial features. In contrast, SpatAttn

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TABLE IX

ABLATION STUDY ON FREQUENCY, SPATIAL, AND FUSION STRATEGIES IN AFSIATTENTION

Method Variant	mIoU (%)	mF1 (%)	Params(M)	FLOPs(G)
Baseline (no attention)	83.93	91.05	58.65	71.39
+ FreqAttn	83.75	90.95	68.43	73.54
+ SpatAttn	84.33	91.31	68.31	73.55
+ Freq + Spat (Concat)	83.64	90.86	68.45	73.60
+ Freq + Spat (Add)	84.53	91.43	68.43	73.56
+ Freq + Spat (Adaptive, Ours)	**85.74**	**92.15**	68.43	73.56

TABLE X

PER-STAGE ADAPTIVE WEIGHTS OF FREQUENCY AND SPATIAL BRANCHES ACROSS DATASETS

Stage	Module	Vaihingen		Potsdam		LoveDA	
		Freq	Spatial	Freq	Spatial	Freq	Spatial
Encoder-1	BADAM	0.6221	0.3779	0.6237	0.3763	0.6221	0.3779
	LWAM	0.6213	0.3787	0.6210	0.3790	0.6225	0.3775
Encoder-2	BADAM	0.6214	0.3786	0.6194	0.3806	0.6213	0.3787
	LWAM	0.6225	0.3775	0.6254	0.3746	0.6230	0.3770
Encoder-3	BADAM	0.6208	0.3792	0.6207	0.3793	0.6202	0.3798
	LWAM	0.6221	0.3779	0.6200	0.3800	0.6224	0.3776
Encoder-4	BADAM	0.6201	0.3799	0.6199	0.3801	0.6213	0.3787
	LWAM	0.6210	0.3790	0.6211	0.3789	0.6220	0.3780
Decoder-1	BADAM	0.6265	0.3735	0.6595	0.3405	0.6281	0.3719
	LWAM	0.6103	0.3897	0.6030	0.3970	0.5977	0.4023
Decoder-2	BADAM	0.6213	0.3787	0.6368	0.3632	0.6156	0.3844
	LWAM	0.5840	0.4160	0.5022	0.4978	0.5890	0.4110
Decoder-3	BADAM	0.5731	0.4269	0.5960	0.4040	0.6040	0.3960

is better at modeling global contextual relationships, which brings certain performance improvements (mIoU increases to 84.33%). Further comparison of the fusion methods of the two types of attention revealed that simple feature concatenation (Concat) failed to effectively integrate the two types of information but instead caused performance degradation (mIoU dropped to 83.64%), while the feature addition (Add) strategy achieved better fusion effect (mIoU increased to 84.53%). In contrast, the adaptive fusion strategy (Adaptive) we proposed can significantly improve the model performance with almost no increase in the number of parameters and computational overhead. The mIoU reaches 85.74% and the mF1 is increased to 92.15%, showing a good balance between performance and efficiency. In Table X, we quantitatively analyze the contribution of frequency (Freq) and spatial weights across different network stages on various datasets. First, frequency-domain information plays a role at every stage of the network, and its contribution varies across different stages and input types. Lower-level features tend to require more frequency information, and this demand changes depending on the input. This demonstrates the necessity of adaptive frequency–spatial interaction, as well as the importance of progressively extracting frequency-domain information throughout the network. Fig. 12 presents the visualization of frequency attention, SpatAttn, and their interaction-based fused attention maps for different input samples.

4) Effectiveness of BiSEM Block: To evaluate the effectiveness of the BiSEM module in reducing information redundancy during feature fusion, we further investigate the

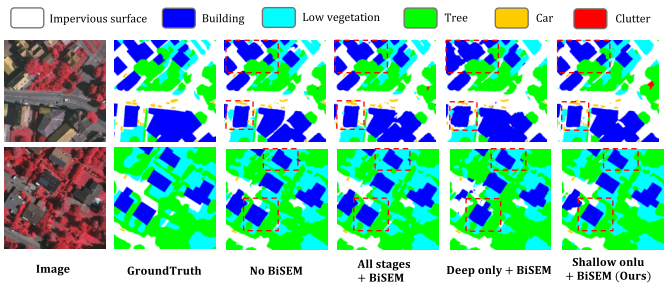


Fig. 13. Visualization of ablation experiments at different locations using BiSEM.

TABLE XI

ABLATION STUDY ON BISEM PLACEMENT

BiSEM Placement	mIoU (%)	mF1 (%)	Params(M)	FLOPs(G)
None	84.72	91.55	68.21	71.05
Deep only	82.76	90.37	71.08	73.22
All stages	84.37	91.32	72.06	78.03
Shallow only (Ours)	**85.74**	**92.15**	**68.43**	**73.65**

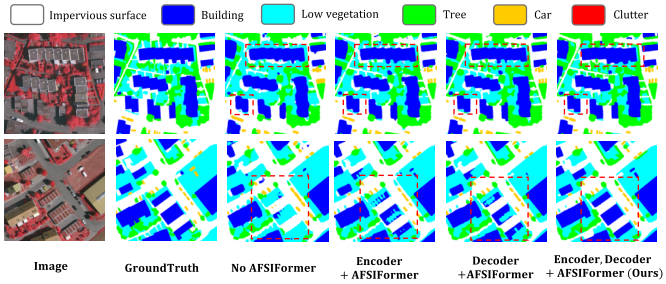


Fig. 14. Visualization of AFSIFormer ablation experiments at different stages of the network.

impact of BiSEM at different network stages in Table XI. Experimental results demonstrate that when BiSEM is applied to the deep layers, although it introduces additional parameters and computational overhead (Params increase to 71.08 M and FLOPs increase to 73.22 G), the performance actually deteriorates (mIoU drops to 82.76% and mF1 drops to 90.37%). This indicates that the semantic gap between deep-layer features is relatively small, and introducing BiSEM at this stage tends to generate information redundancy. When BiSEM is applied to all stages, the performance shows slight improvement (mIoU increases to 84.37% and mF1 increases to 91.32%), but the computational overhead increases significantly (FLOPs increase to 78.03 G), leading to reduced efficiency. In contrast, our method introduces BiSEM only at the shallow layers, achieving an optimal balance between performance and efficiency, with mIoU improved to 85.74% and mF1 reaching 92.15%, while maintaining parameter count and computational complexity nearly identical to the baseline (68.43 M, 73.65 G). This demonstrates that there exists a substantial semantic gap between shallow and deep features, and BiSEM can effectively mitigate this semantic disparity, facilitating cross-level semantic fusion and thereby enhancing model performance without significantly increasing computational overhead. The visualization results in Fig. 13 further validate

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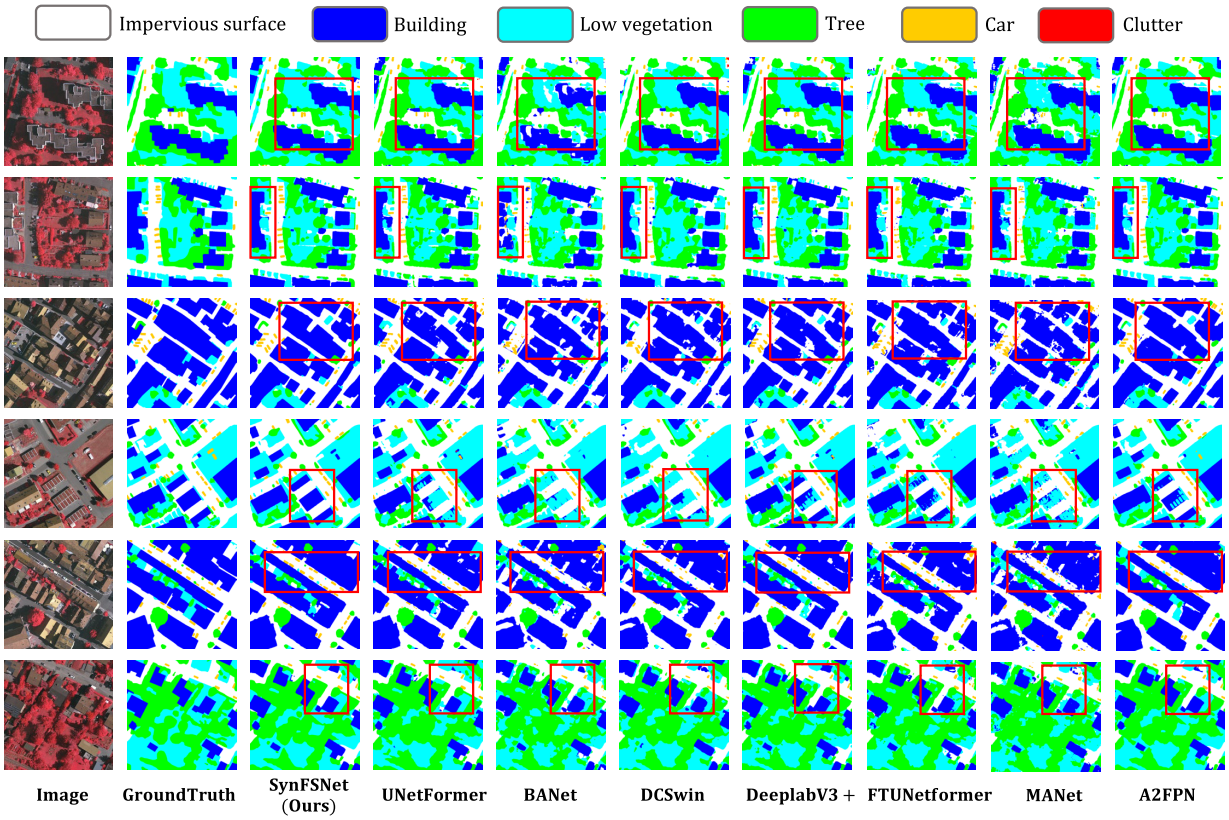


Fig. 15. Qualitative comparisons between ours and other models on the ISPRS Vaihingen dataset. The red box highlights the improvement in performance achieved by our model compared to others.

this analysis, showing that the configuration with BiSEM introduced only at shallow layers not only achieves excellent segmentation performance in dense areas, such as buildings and trees, but also enables fine-grained segmentation of small objects like cars, confirming the effectiveness of the module design.

5) Impact of AFSIFormer Integration at Different Network Stages: In Table XII, we explore the impact of placing the AFSIFormer module at different positions within the network. Experiments show that when AFSIFormer is applied only to the encoder, the mIoU increases to 84.45% and the mF1 reaches 91.37%. When applied only to the decoder, the mIoU rises to 84.47% and the mF1 to 91.39%. Both settings yield improvements compared to the baseline (mIoU of 83.93% and mF1 of 91.05%), but the improvement is limited. In contrast, when we deployed AFSIFormer in both the encoder and decoder (i.e., the all stages), the model achieved the best results, with mIoU reaching 85.74% and mF1 increasing to 92.15%. This shows that synergy modeling of global and local information in the full network stage can achieve more sufficient information flow and semantic fusion, significantly enhancing the segmentation performance on aerial images. Fig. 14 shows the visualization results of the ablation experiment. When AFSIFormer is only applied to the encoder or decoder, the model will lose global information and frequency-domain information to varying degrees, resulting in poor segmentation of categories, such as building and tree, that are highly dependent on global context. After AFSIFormer is introduced in the encoder and decoder stages, the overall segmentation effect of the model is significantly improved.

TABLE XII
ABLATION STUDY OF AFSIFORMER PLACEMENT AT DIFFERENT NETWORK STAGES

Method	Encoder	Decoder	mIoU(%)	mF1(%)	Params(M)	FLOPs(G)
AFSIFormer+	✗	✗	83.93	91.05	58.15	71.39
	✓	✗	84.45	91.37	66.47	72.61
	✗	✓	84.47	91.39	60.60	72.34
Ours	✓	✓	**85.74**	**92.15**	68.43	73.56

F. Visualization of Experimental Results

To further validate the effectiveness of our method, Fig. 15 presents a comparison of the prediction results between our model and current mainstream CNN–transformer-based models as well as classic models on the ISPRS Vaihingen dataset. This dataset presents various challenges, including densely distributed buildings, small objects, such as cars, and trees with blurred boundaries. Due to the introduction of the AFSIFormer, our model shows stronger feature modeling capabilities in the fuzzy boundary areas and can accurately extract edge features of categories, such as tree and low vegetation, significantly outperforming other comparison models. Through the block-level residual coupling mechanism enhanced by feature complementarity, global and local

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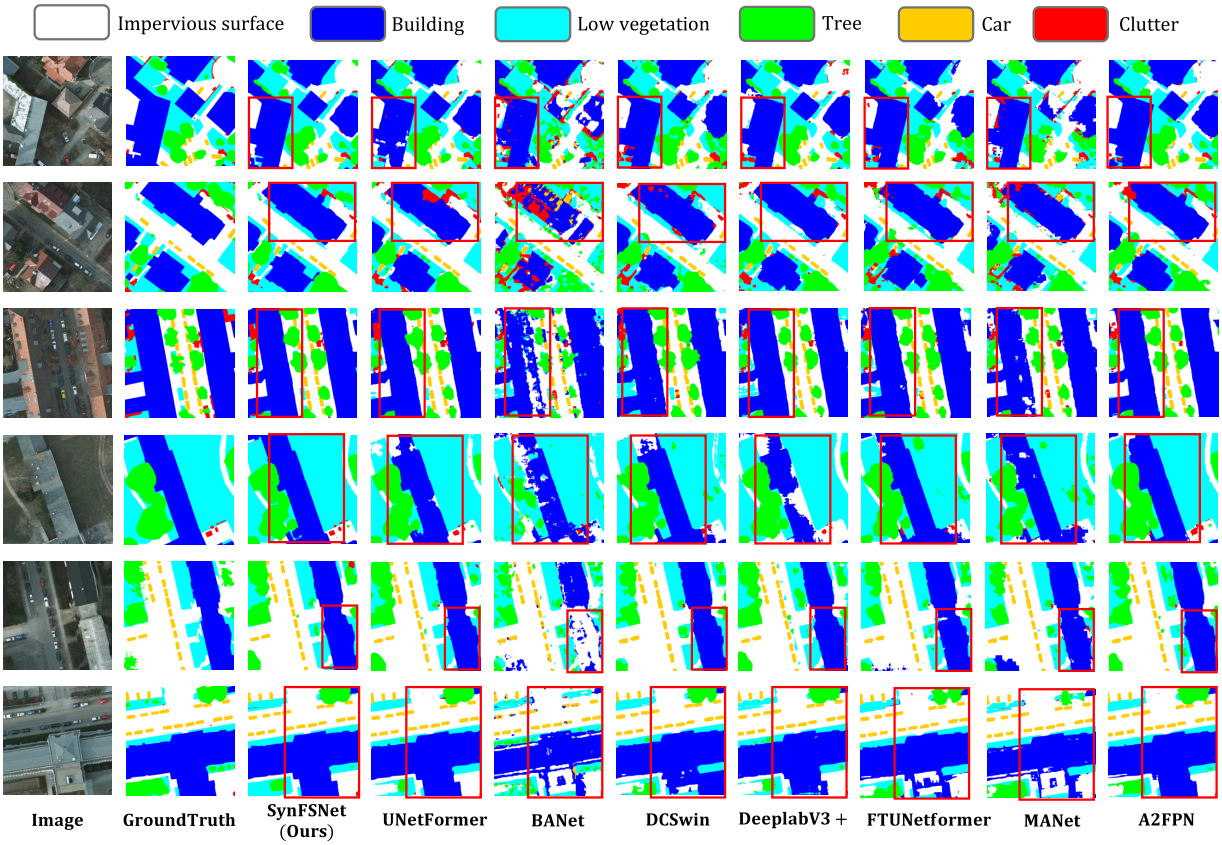


Fig. 16. Qualitative comparisons between ours and other models on the ISPRS Potsdam dataset. The red box highlights the improvement in performance achieved by our model compared to others.

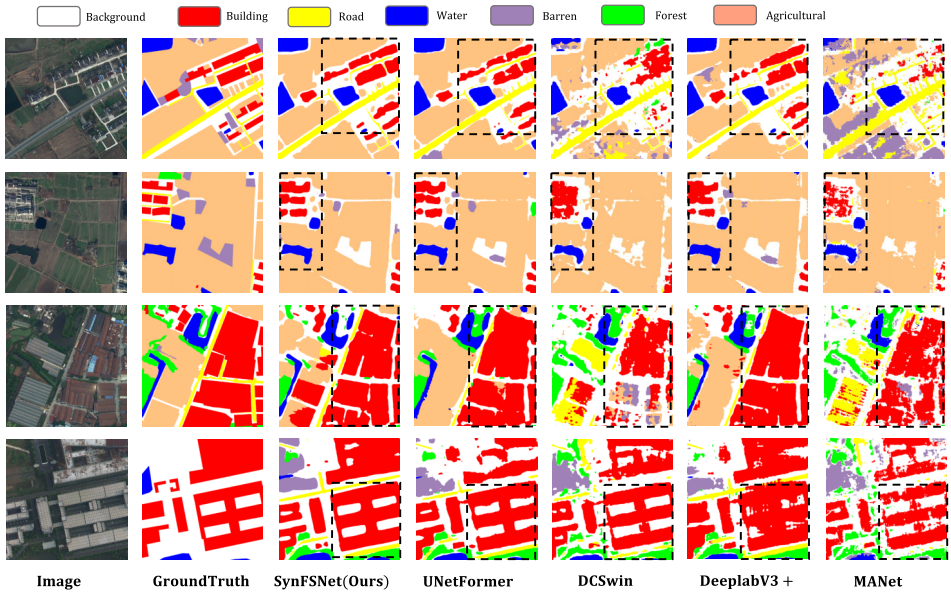


Fig. 17. Qualitative comparisons between ours and other models on the LoveDA dataset. The black box highlights the improvement in performance achieved by our model compared to others.

information runs through the entire network structure. Compared to UNetFormer, which integrates local and global information only during the decoding stage, our method achieves effective coordination during both the encoding and decoding processes, thereby exhibiting superior performance in segmenting large-scale variations, such as buildings, and small objects, such as cars. Furthermore, the BiSEM module enhances the segmentation accuracy of small objects, such as cars, by finely integrating low-level and high-level semantic features, achieving optimal performance.

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The comparison results with other advanced models on the ISPRS Potsdam test set are shown in Fig. 16. This dataset shares similar category composition and challenges with the Vaihingen dataset. Our method consistently demonstrates superior segmentation performance and achieves strong results across multiple target categories.

The LoveDA dataset covers both urban and rural areas, with challenges of complex objects and inconsistent categories. Since the LoveDA dataset does not have a public test set label, Fig. 17 shows the prediction results of our model compared with other advanced models on the LoveDA validation set. The first two rows are the comparison results for rural images, and the last two rows are the comparison results for urban images. It can be seen that when facing areas with dense buildings, roads, and water bodies, our method can better maintain boundary continuity and regional integrity and achieve the best segmentation effect. Our model also shows stronger adaptability and generalization when dealing with categories, such as large-scale buildings, water bodies with complex morphology, and forests. Overall, our proposed method demonstrates excellent segmentation performance in multiple aerial scenarios and under different challenging conditions.

VI. Conclusion

This article proposes an efficient aerial image semantic segmentation network, SynFSNet, which integrates AFSIFormer, BiSEM, and a block-level residual coupling architecture. It achieves a fundamental shift in the utilization of frequency-domain information—from external fusion to internal integration, from staged usage to continuous modeling, and from result-level enhancement to process-level synergy. AFSIFormer employs AFSIAttention to enable dynamic fine-grained frequency–spatial interactions and leverages progressive learning strategies, BADAM and LWAM, to effectively reduce computational overhead while maintaining segmentation quality. The block-level residual coupling architecture enables continuous propagation of frequency and global information without relying on a complex global modeling backbone. Meanwhile, BiSEM addresses the challenges of feature fusion. This integrated approach maximizes the potential of frequency-domain information in remote sensing semantic segmentation without significantly increasing computational cost, achieving a synergy between efficiency and performance. Extensive experiments conducted on the Vaihingen, Potsdam, and LoveDA datasets demonstrate that SynFSNet outperforms the existing SOTA methods in terms of segmentation accuracy and generalization capability. Comprehensive ablation studies further validate the effectiveness of each proposed component.

References

[1] R. Li, S. Zheng, C. Duan, L. Wang, and C. Zhang, "Land cover classification from remote sensing images based on multi-scale fully convolutional network," *Geo-Spatial Inf. Sci.*, vol. 25, no. 2, pp. 278–294, Apr. 2022.

[2] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[3] Y. He, J. Wang, Y. Zhang, and C. Liao, "An efficient urban flood mapping framework towards disaster response driven by weakly supervised semantic segmentation with decoupled training samples," *ISPRS J. Photogramm. Remote Sens.*, vol. 207, pp. 338–358, Jan. 2024.

[4] W. Bo, J. Liu, X. Fan, T. Tjahjadi, Q. Ye, and L. Fu, "BASNet: Burned area segmentation network for real-time detection of damage maps in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627913.

[5] F. Chen, H. Balzter, F. Zhou, P. Ren, and H. Zhou, "DGNet: Distribution guided efficient learning for oil spill image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4201317.

[6] R. Lian, W. Wang, N. Mustafa, and L. Huang, "Road extraction methods in high-resolution remote sensing images: A comprehensive review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5489–5507, 2020.

[7] P. Shamsolmoali, M. Zareapoor, H. Zhou, R. Wang, and J. Yang, "Road segmentation for remote sensing images using adversarial spatial pyramid networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4673–4688, Jun. 2021.

[8] D. Griffiths and J. Boehm, "Improving public data for building segmentation from convolutional neural networks (CNNs) for fused airborne LiDAR and image data using active contours," *ISPRS J. Photogramm. Remote Sens.*, vol. 154, pp. 70–83, Aug. 2019.

[9] R. Cao et al., "Deep learning-based remote and social sensing data fusion for urban region function recognition," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 82–97, May 2020.

[10] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5607713.

[11] H. Shen et al., "Missing information reconstruction of remote sensing data: A technical review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 61–85, Sep. 2015.

[12] B. Zhao, Y. Zhong, and L. Zhang, "A spectral–structural bag-of-features scene classifier for very high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 116, pp. 73–85, Jun. 2016.

[13] Y. Chen et al., "High-resolution remote sensing image change detection based on Fourier feature interaction and multiscale perception," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5539115.

[14] J. Zhang, Y. Rao, X. Huang, G. Li, X. Zhou, and D. Zeng, "Frequency-aware multi-modal fine-tuning for few-shot open-set remote sensing scene classification," *IEEE Trans. Multimedia*, vol. 26, pp. 7823–7837, 2024.

[15] G. Strang, "The discrete cosine transform," *SIAM Rev.*, vol. 41, no. 1, pp. 135–147, 1999.

[16] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[17] S. Zheng, Z. Wu, Y. Xu, Z. Wei, and A. Plaza, "Learning orientation information from frequency-domain for oriented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628512.

[18] L. Chen, L. Gu, D. Zheng, and Y. Fu, "Frequency-adaptive dilated convolution for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3414–3425.

[19] X. Li, F. Xu, H. Gao, F. Liu, and X. Lyu, "A frequency domain feature-guided network for semantic segmentation of remote sensing images," *IEEE Signal Process. Lett.*, vol. 31, pp. 1369–1373, 2024.

[20] J. Liu, D. Zhang, L. He, X. Yu, and W. Han, "MFAGNet: Multi-scale frequency attention gating network for land cover classification," *Int. J. Remote Sens.*, vol. 44, no. 21, pp. 6670–6697, Nov. 2023.

[21] X. Li et al., "SSCNet: A spectrum-space collaborative network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 15, no. 23, p. 5610, Dec. 2023.

[22] G. Wei, J. Xu, W. Yan, Q. Chong, H. Xing, and M. Ni, "Dual-domain fusion network based on wavelet frequency decomposition and fuzzy spatial constraint for remote sensing image segmentation," *Remote Sens.*, vol. 16, no. 19, p. 3594, Sep. 2024.

[23] H. Zhang, G. Xie, L. Li, X. Xie, and J. Ren, "Frequency-domain guided Swin transformer and global–local feature integration for remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612611.

[24] P. Wang et al., "MT_GAN: A SAR-to-optical image translation method for cloud removal," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 180–195, Jul. 2025.

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[25] P. Wang, Z. He, B. Huang, M. Dalla Mura, H. Leung, and J. Chanussot, "VOGTNet: Variational optimization-guided two-stage network for multispectral and panchromatic image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9268–9282, May 2025.

[26] Y. Yang, G. Yuan, and J. Li, "SFFNet: A wavelet-based spatial and frequency domain fusion network for remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3000617.

[27] T. Zhang and R. P. Dick, "Spatial-frequency network for segmentation of remote sensing images," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2023, pp. 3553–3557.

[28] J. Fan, J. Li, Y. Liu, and F. Zhang, "Frequency-aware robust multidimensional information fusion framework for remote sensing image segmentation," *Eng. Appl. Artif. Intell.*, vol. 129, Mar. 2024, Art. no. 107638.

[29] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[30] C. Guo, B. Fan, Q. Zhang, S. Xiang, and C. Pan, "AugFPN: Improving multi-scale feature learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12595–12604.

[31] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[32] L. Fan, Y. Zhou, H. Liu, Y. Li, and D. Cao, "Combining Swin transformer with UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5530111.

[33] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.

[34] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.

[35] R. Zhang, Q. Zhang, and G. Zhang, "LSRFormer: Efficient transformer supply convolutional neural networks with global information for aerial image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610713.

[36] P. Wang et al., "Low-rank tensor completion pansharpening based on haze correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5405720.

[37] P. Wang et al., "CIRSM-net: A cyclic registration network for SAR and optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5610619.

[38] J. Liu, Z. Wang, F. Luo, T. Guo, F. Yang, and X. Gao, "ESMS-net: Enhancing semantic-mask segmentation network with pyramid atrousformer for remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5650414.

[39] G. P. O. Reddy, "Digital image processing: Principles and applications," in *Geospatial Technologies in Land Resources Mapping, Monitoring and Management*, G. P. O. Reddy and S. K. Singh, Eds., Cham, Switzerland: Springer, 2018, pp. 101–126, doi: 10.1007/978-3-319-78711-4_6.

[40] D. Yin, R. G. Lopes, J. Shlens, E. D. Cubuk, and J. Gilmer, "A Fourier perspective on model robustness in computer vision," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 13255–13265.

[41] R. Suvorov et al., "Resolution-robust large mask inpainting with Fourier convolutions," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 2149–2159.

[42] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11534–11542.

[43] I. Kotaridis and M. Lazaridou, "Remote sensing image segmentation advances: A meta-analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 309–322, Mar. 2021.

[44] Y. Bazi, L. Bashmal, M. M. A. Rahhal, R. A. Dayil, and N. A. Ajlan, "Vision transformers for remote sensing image classification," *Remote Sens.*, vol. 13, no. 3, p. 516, Feb. 2021.

[45] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[46] X. Ma, X. Zhang, and M.-O. Pun, "RS^{3}Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, 2024, Art. no. 6011405.

[47] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6506105.

[48] L. Gao et al., "STransFuse: Fusing Swin transformer and convolutional neural network for remote sensing image semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10990–11003, 2021.

[49] M. Yu, L. He, Z. Shen, and M. Lv, "STRD-net: A dual-encoder semantic segmentation network for urban green space extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4413913.

[50] Y. Chen, Y. Wang, S. Xiong, X. Lu, X. X. Zhu, and L. Mou, "Integrating detailed features and global contexts for semantic segmentation in ultrahigh-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703914.

[51] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[52] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[53] F. Rottensteiner et al., "International society for photogrammetry and remote sensing, 2D semantic labeling contest," Int. Soc. Photogramm. Remote Sens. (ISPRS), Inst. Photogramm. GeoInf. (IPI), Leibniz Univ. Hannover, Hannover, Germany, Tech. Rep. 29, 2020. Accessed: Aug. 17, 2025. [Online]. Available: https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/semantic-labeling.aspx

[54] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[55] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[56] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, Munich, Germany. Cham, Switzerland: Springer, 2015, pp. 234–241.

[57] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[58] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[59] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[60] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiseNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 325–341.

[61] R. P. K. Poudel, S. Liwicki, and R. Cipolla, "Fast-SCNN: Fast semantic segmentation network," 2019, *arXiv:1902.04502*.

[62] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7262–7272.

[63] L. Wang, R. Li, D. Wang, C. Duan, T. Wang, and X. Meng, "Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images," *Remote Sens.*, vol. 13, no. 16, p. 3065, Aug. 2021.

[64] R. Li, L. Wang, C. Zhang, C. Duan, and S. Zheng, "A2-FPN for semantic segmentation of fine-resolution remotely sensed images," *Int. J. Remote Sens.*, vol. 43, no. 3, pp. 1131–1155, Feb. 2022.

[65] X. Zhang, Z. Weng, P. Zhu, X. Han, J. Zhu, and L. Jiao, "ESDINet: Efficient shallow-deep interaction network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607615.

[66] Y. Ni, J. Liu, W. Chi, X. Wang, and D. Li, "CGGLNet: Semantic segmentation network for remote sensing images based on category-guided global–local feature interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5615617.

[67] D. Yu and S. Ji, "Long-range correlation supervision for land-cover classification from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4409814.

[68] Y. Chen et al., "SparseFormer: A credible dual-CNN expert-guided transformer for remote sensing image segmentation with sparse point annotation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5607316.

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[69] J. Zhang, M. Shao, Y. Wan, L. Meng, X. Cao, and S. Wang, "Boundary-aware spatial and frequency dual-domain transformer for remote sensing urban images segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5637718.
[70] Y. Zhou, J. Huang, C. Wang, L. Song, and G. Yang, "Xnet: Wavelet-based low and high frequency fusion networks for fully-and semi-supervised semantic segmentation of biomedical images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 21085–21096.
[71] F. Gao, M. Fu, J. Cao, J. Dong, and Q. Du, "Adaptive frequency enhancement network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5619415.
[72] H. Cai, J. Li, M. Hu, C. Gan, and S. Han, "EfficientViT: Lightweight multi-scale attention for high-resolution dense prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 17302–17313.
[73] Y. Ji et al., "DDP: Diffusion model for dense visual prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 21684–21695.
[74] R. Xu, C. Wang, J. Zhang, S. Xu, W. Meng, and X. Zhang, "RSSFormer: Foreground saliency enhancement for remote sensing land-cover segmentation," *IEEE Trans. Image Process.*, vol. 32, pp. 1052–1064, 2023.
[75] X. Ma et al., "LOGCAN++: Adaptive local–global class-aware network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4404216.
[76] L. Zhu, X. Wang, Z. Ke, W. Zhang, and R. Lau, "BiFormer: Vision transformer with bi-level routing attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10323–10333.
[77] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
[78] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[79] Z. Liu et al., "Swin transformer V2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12009–12019.



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